

KINETIX VIDEO COACH

Quantitative Biomechanical Review Assistant for Fast Bowling Technique

Product Type: Video Review Assistant

Kinematic Scope: 2 Sagittal Metrics

Camera Setup: 2.8m–3.2m Net Clearance
Stencil

Validation Status: Stage 1 Active (18 Bowlers)

1. PRODUCT VISION & EPISTEMIC SEPARATION

Adolescent fast bowlers suffer high rates of lumbar bone-stress injuries (~16% seasonal prevalence, ~28% lifetime prevalence). While high-performance centers utilize expensive multi-camera marker laboratories, grassroots clubs rely entirely on subjective eyesight.

EXPLICIT BOUNDARY: NON-DIAGNOSTIC COACHING TOOL (ZERO INJURY PREDICTIONS)

Kinetix is a quantitative, biomechanics-informed video-review assistant, not an injury predictor or medical diagnostic. It extracts 2D skeletal coordinates from ordinary high-frame-rate smartphone video, benchmarks them against personal reference and rolling baselines, and flags *meaningful mechanical deviations*. It strictly enforces the epistemic chain: **Observed** (pixel angles) → **Inferred** (baseline delta ± uncertainty) → **Interpreted** (candidate technical pattern). Final coaching and workload determinations remain exclusively with the accredited coach.

2. PHYSICAL & OPTICAL NET LANE CONSTRAINTS

1. Geometry & Stencil

- Camera distance: 2.8m–3.2m (fits 3.0m net lanes).
- Tripod fixed at 1.1m (bowler hip level).
- Crease locked to in-app stencil (roll/pitch < 3.0°).

2. Net Mesh Clearance

- Lens placed 15cm flush against net netting.
- Eliminates net wire occlusion over lower limb.
- Wide lens (0.6x/1x) with LUT un-warp.

3. Lighting & Thermal Guard

- Natural daylight capture (8:00 AM – 4:30 PM).
- Shutter locked ≤ 1/1000s (no motion blur).
- Ingestion audit rejects files with frame drops.

3. DECOUPLED TRI-LAYER ARCHITECTURE

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| LAYER 1: MEASUREMENT ENGINE (Vision & Signal Processing) |
| Smartphone Video (60/120 FPS) --> Hardware Audit: Delta t Continuity (Rejects thermal throttling) |
| --> RTMDet-nano + RTMPose-s / MediaPipe Edge Extraction |
| --> Quality Firewall: Suppresses analysis if p_knee < 0.70 or p_hip < 0.70 (e.g. net mesh wire occlusion) |
| --> Forward-Backward 4th-Order Butterworth Filter (Zero-phase, f_c calibrated against ground truth) |
| --> Multi-Cue FFS Event Detector: Fused ankle height minima + horizontal decel spike + velocity zero-crossing |
| --> OUTPUT: Calibrated 2D Joint Coordinates [x, y], FFS Frame Index, Landmark Confidence (p_i) |
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| LAYER 2: INTERPRETATION ENGINE (Statistical Triangulation) |
| 1. Dual-Baseline Evaluation: |
|   - Coach-Confirmed Fixed Reference Baseline: Median + IQR across 8-10 un-fatigued benchmark deliveries |
|   - Rolling 6-Week Median: Captures macro longitudinal adaptation across sessions |
| 2. Rolling Window Anomaly Filter: |
|   - Single Ball Deviation: Labeled "Unclassified Deviation" (Yellow Replay Flag; no alarm) |
|   - >= 3 of 5 Balls Deviation: Asserts "Technical Concern" flag (Directional consistency confirmed) |
| 3. Uncertainty Attribution: Error band labeled "Pending Stage 1 Calibration" (MAE ± 1.96 * SD_residual) |
| --> OUTPUT: Categorical Status (Form Benchmark | Mechanical Watch | Technical Concern | Data Suppressed) |
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| LAYER 3: COACHING ENGINE (Human-in-the-Loop & Actions) |
| 1. Coach Action Menu: [Approve Drill] | [Dismiss: Tactical Yorker / Slower Ball / Slip] | [Nudge FFS Frame ±1] |
| 2. Mandatory Clinical Override: Pain, structural discomfort, or medical restriction strictly overrides prompts |
| 3. Curated Remedial Library: Credentialed S&C exercises (UKCC Level 3 / BCCI Level 2 authored; zero LLM drift) |
| 4. Ephemeral Video Storage: Raw video deleted after coordinate extraction; privacy-safe WhatsApp cards |
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4. THE SCOPED 2-METRIC KINEMATIC FOUNDATION

CANDIDATE METRIC	OPTICAL PLANE & VIEW	DIRECT OBSERVATION	INFERRED MECHANICAL ASSESSMENT	SPORTS SCIENCE LITERATURE CONTEXT
Front Knee Extension at FFS (Vertex: Knee; Rays: Hip, Ankle)	Sagittal Crease View (Side-on, 2.8m–3.2m distance)	2D interior angle at the exact multi-cue frame of front foot strike (FFS).	Evaluates whether knee maintains baseline extension (±SD) or exhibits acute flexion (>1.5 SD below median).	A braced front knee acts as a rigid mechanical lever, converting linear approach momentum into angular torso speed. A flexing knee absorbs forces through muscular work, reducing braking efficiency.
Forward Trunk Tilt at Release (Vertex: Mid-Pelvis; Spine vs. Vertical)	Sagittal Crease View (Side-on delivery stride)	Forward inclination of the thoracic spine vector relative to vertical pitch normal at release.	Detects acute within-spell drift (>4° from Spell Over 1) or deviation from personal reference baselines.	Forward trunk lean directly influences release height, ball trajectory, and forward momentum transfer. Rapid within-spell drift indicates kinetic chain fatigue.

5. DUAL-BASELINE ASSESSMENT & "WHY WAS THIS FLAGGED?" IN-APP CARD

OBSERVED KINEMATIC RECORD	BASELINE COMPARISON	UNCERTAINTY ASSESSMENT	ENGINE STATUS & HUMAN-IN-THE-LOOP ACTION
Front Knee: 134.0° (Confidence: 94% Frame: #73)	-14.0° vs. Fixed Ref (148.0°) -11.0° vs. Rolling Median	Error Band: Pending Stage 1 Calibration (MAE ± 1.96 * SD_res). Material deviation.	TECHNICAL CONCERN (3 of last 5 balls flagged) Interpretation: Meaningful deviation from baseline front-leg bracing. Drill Proposed: Step-Down Landing Holds (3 × 6 reps). <i>Coach Action:</i> [Approve Drill] [Dismiss: Tactical Delivery] [Nudge FFS]
Trunk Lean: 21.2° (Confidence: 91% Frame: #88)	+3.8° drift from Over 1 baseline (17.4°)	Within estimated noise band. Evaluated as non-critical trend.	MECHANICAL WATCH (Isolated occurrence) Interpretation: Possible within-spell trunk drift (flagged for video replay). <i>Coach Action:</i> Logged to session history. No corrective drill triggered.
Front Knee: 141.0° (Confidence: 41% Frame: #74)	N/A (Confidence gated).	Suppressed by Quality Firewall (p_i < 0.70).	DATA SUPPRESSED Reason: Landmark visibility below threshold (netting wire occlusion). <i>Action:</i> Adjust tripod 15cm flush against net mesh. Re-record.

"WHY WAS THIS FLAGGED?" SCREEN TRANSPARENCY CARD (IN-APP UX)

Observed Delivery: Ball #24 (Over 4.2) • **Front Knee Angle:** 134.0° • **Landmark Confidence:** 94%
Fixed Reference Baseline: 148.0° (IQR: 146.0°–151.0°, Confirmed 2026-08-10) • **Observed Delta:** -14.0°
Measurement Uncertainty Band: Pending Stage 1 Calibration (MAE ± 1.96 * SD_residual)
Trigger Context: 3 of the last 5 deliveries exhibited consistent flexion under impact (Ball #20: 135°, Ball #22: 133°, Ball #24: 134°).
Biomechanical Interpretation: Persistent deviation from athlete's established bracing mechanics; indicates ground force absorption deficit.
Proposed Remedial Drill: Step-Down Landing Holds (3 × 6 reps off 12-inch box; focus on isometric knee extension).
Coach Action Controls: [Approve & Assign Drill] • [Dismiss: Tactical Yorker / Slower Ball] • [Nudge FFS Frame ±1]
Disclaimer: This software is a quantitative coaching tool, not an orthopedic or injury diagnostic. Any pain overrides automated suggestions.

6. EMPIRICAL STAGE 1 VALIDATION PROTOCOL & TARGET SLAS

EXPERIMENTAL PARAMETER / TARGET SLA	DESIGN TARGET & VERIFICATION PROTOCOL	MEASURED REALITY (TRIAL LOG)
Cohort Size & Diversity	18 Distinct Bowlers (12 Calibration / 6 Strictly Held-Out Unseen Test Set).	[Bowlers]
Total Deliveries Analyzed	150+ Valid Deliveries recorded outdoors at 60 or 120 FPS.	[Deliveries]
Ground Truth Benchmark	Kinovea manual frame-by-frame joint angle annotation by accredited biomechanist.	[Biomechanist Name:]
Front Knee Angle Accuracy	MAE ≤ 4.0° on the 6 held-out bowlers (not overfitted to calibration set).	[° MAE]
Forward Trunk Tilt Accuracy	MAE ≤ 3.5° on the 6 held-out bowlers.	[° MAE]
FFS Event Timing Accuracy	≤ 33 ms at 60 FPS / ≤ 17 ms at 120 FPS (≤ ± 2 frames).	[ms / frames]
Classification Concordance	Decision Concordance > 80% with accredited coach judgment.	[% Agreement]
Quality Firewall Rejection Rate	< 15% of clips rejected due to camera motion, shutter blur, or occlusion.	[% Rejection]
Coach Dismissal Rate (False Burden)	Target < 25% of flagged concerns dismissed by coach across 30 deliveries.	[% Dismissed]
Processing Latency SLA	Target P50 ≤ 8s P95 ≤ 15s per delivery stride on standard mobile network.	[P50: s P95: s]
Field Setup SLA	Target Setup ≤ 30s using crease alignment viewfinder stencil.	[s Setup]
Coach Retention (Pilot Target)	Target > 65% of coaches running ≥ 2 analysis sessions per week across 4 weeks.	[% Retention]

7. MINOR PRIVACY, GOVERNANCE & ETHICAL GUARDRAILS

- **Digital Parental Consent Gate:** Adolescent fast bowlers (< 18 years) require verifiable digital guardian acknowledgment before profile activation.
- **Ephemeral Processing Pipeline:** Raw high-speed video files are processed in volatile memory and purged immediately upon keypoint coordinate extraction. Only anonymous numerical vector streams ([x, y, p_i, t]) are retained for longitudinal tracking.
- **Closed WhatsApp Coach-Guardian Channel:** Exported summary cards contain strictly validated angles, trend charts, and approved drill recommendations—raw video clips are never transmitted or hosted on public web links.
- **Strict Scope Boundary:** No medical diagnostic claims, no injury probability calculations, and zero autonomous exercise invention. Pain or medical restrictions unconditionally override software suggestions.

8. DEFINITIVE 3-MINUTE HACKATHON DEMO SCRIPT

1. **The Problem & Grounded Positioning (0:00–0:40):** "Adolescent fast bowlers face severe skeletal loading, yet grassroots academies evaluate 130 km/h delivery strides using only eyesight. We didn't build an unvalidated 'AI injury doctor.' We built Kinetix: an AI-assisted video coaching assistant that tracks quantitative kinematic baselines and detects meaningful mechanical drift using ordinary smartphone cameras."
2. **The Live Proof & Rolling Window (0:40–1:50):** Run the tool against calibrated deliveries of the same bowler:
 - *Delivery A (Benchmark Form):* Knee braces cleanly at 167.0° → UI renders **Form Benchmark** with keypoint confidence markers.
 - *Deliveries B1–B3 (Rolling Window Drift):* Demonstrates that an isolated collapse generates an **Unclassified Deviation** yellow replay flag, but 3 consecutive flexed strides trigger **Technical Concern**, presents the *Why Flagged* transparency card, and prompts coach review: [Approve Step-Down Drill] [Dismiss: Tactical Yorker] [Nudge FFS].
 - *The Trust Slide:* Play an occluded net clip where the app transparently outputs: DATA SUPPRESSED: Insufficient Landmark Visibility rather than hallucinating an angle.

3. **Engineering Integrity & Validation Roadmap (1:50–3:00):** Walk through the three-layer decoupled architecture, explain how coach accept/dismiss feedback trains our proprietary decision asset, and present the active Stage 1 validation protocol.

9. MASTER REVIEW & DEFENSIBILITY SCORECARD

EVALUATION DIMENSION	RATING	DETAILED PRACTICAL VERDICT & REAL-WORLD JUSTIFICATION
Problem Selection	9.5 / 10	Real, acute pain point in grassroots cricket. High prevalence of youth lumbar stress injuries (~16% seasonal, ~28% lifetime) paired with zero accessible quantitative analysis tools in academies.
Scope Discipline	9.5 / 10	Aggressively scoped down from a bloated "full biomechanics suite" to two candidate sagittal metrics (Front Knee Extension at FFS and Forward Trunk Tilt at Release) captured from a single side-on crease view .
Scientific & Epistemic Defensibility	8.0 / 10	All medical and injury prediction claims are eliminated. Operates strictly as a <i>quantitative video-review assistant</i> . Clearly separates Observed (pixels) from Inferred (baseline delta) from Interpreted (coaching pattern).
Product & UX Architecture	8.5 / 10	Mature design: decoupled tri-layer architecture, 3-of-5 rolling window anomaly filter, explicit confidence suppression ($p_i < 0.70$), and an actionable coach override menu (`Approve`, `Dismiss Tactical`, `Nudge FFS`).
Real-World Field Feasibility	7.2 / 10	Addressed 2.8m net lane clearance and outdoor thermal frame drops. However, outdoor sunlight, camera tripod alignment discipline, and dirty net netting still require careful management by coaches.
Hackathon Winning Potential	8.8 / 10	High presentation impact: live demo using real calibrated deliveries, an intentional occlusion suppression demonstration, and an auditable "Why Was This Flagged?" card build deep judge trust.
Current Empirical Maturity	6.0 / 10	The experimental protocol is rigorous (18 bowlers, 150+ deliveries, athlete-partitioned test splits), but the right-hand column is currently unfilled pending trial execution.
Composite Verdict	8.5 / 10	Strong Prototype • Scoped Implementation Freeze • Stage 1 Validation Active. Ready for physical field testing.